MIND MAP FOR NEET ASPIRANTS

BOTANY

Molecular Basis of Inheritance

(One Shot)

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Molecular Basis of Inheritance

The molecule that controls inheritance is called NUCLEIC ACID / POLYNUCLEOTIDE!

Watson, crick (on the basis of x-ray diffraction data of Wilkins & Franklin) - double helix model

Genetic material ← DNA in majority of organisms

RNA → In a few viruses only.

- $\phi \times 174$: 5386 Nucleotides
- λ-Bacteriophage : 48502 bp
- E.coli : 4.6×10^6 bp
- Human : 3.3×10^9 bp

Meischer (1869) - discovered DNA (named it nuclein).

■ Polynucleotide chain:

Ribose

Deoxyribose (-1'O')

3-Parts: N-base + sugar + phosphate

Purine | Pyrimidine

A, G | C, U, T

N:1,3,7,9

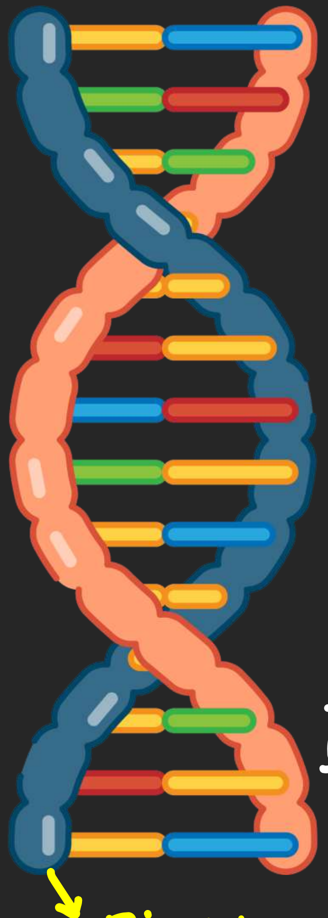
N:1,3

Base + sugar = Nucleoside

Nucleoside + P = Nucleotide

↓ Polymerise

Polynucleotide



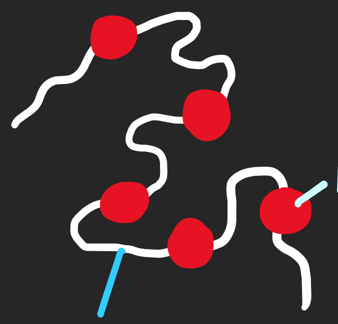
34A° = Pitch (360°) ↓ 10bp

3.4A°

Right hand coiled ; 2 anti || Strands ; A=T & C≡G.

Packaging of DNA Helix: Histones contains basic amino acids (lysine + arginine) ∴ positive

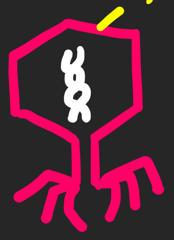
- Dimers of H_2A, H_2B, H_3 & H_4 makes histone octamers: wraps around 200bp DNA around it.

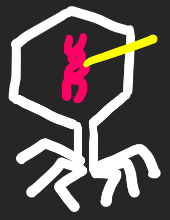


- @higher level of packaging; non-histone chromosomal proteins are required!

- Biochemical characterisation: Avery, McLeod & McCarty proved that DNA is gen. mat. & not RNA or Protein.

- Hershey & chase experiment: unequivocal proof





- Transforming Principle: Griffith (1928) → used Diplococcus → S - virulant; R - non-virulant

S-strain + Mice: Mice die

R-strain + Mice: Survived

Heat killed S-strain: Survived

Heat killed S-strain + R-strain: Died → S-strain recovered from mice heart

- Properties of Genetic material:
  1. Should be able to replicate
  2. Stable
  3. Shall mutate slowly
  4. Express itself

DNA is better!

- Central Dogma by Crick:

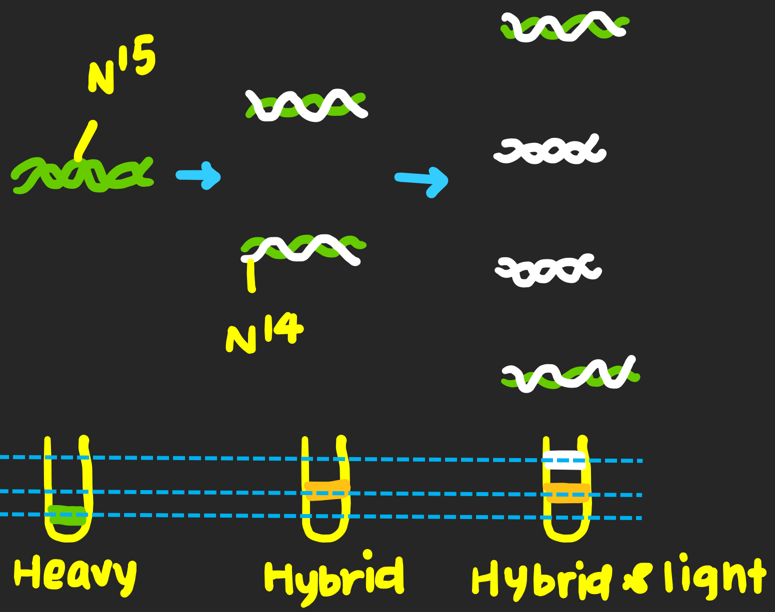
$$DNA \xrightarrow{\text{Transc.}} RNA \xrightarrow{\text{Transl.}} Protein$$

replication

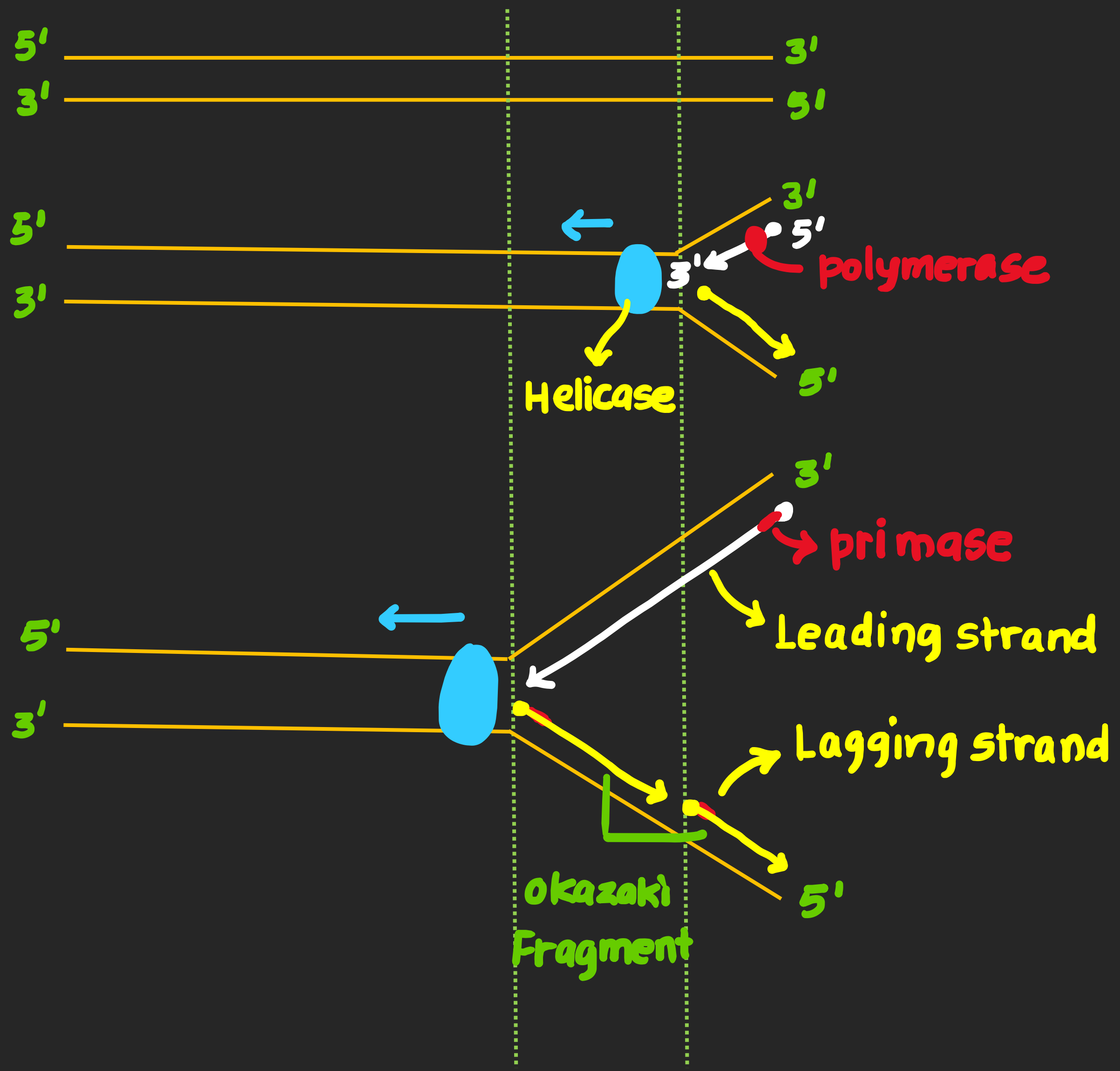
Replication

Semiconservative in nature

- Proved by Meselson & Stahl using N^{15} (heavy isotope).
- By Taylor in Faba beans using radioactive (T)



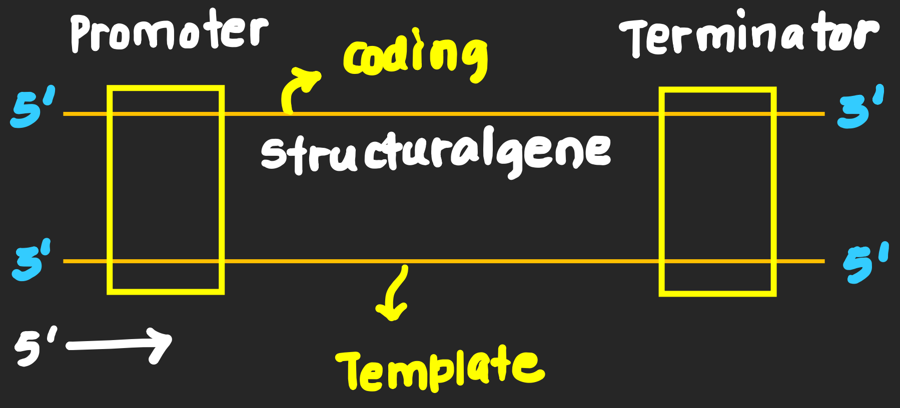
Replication machinary : DNA polymerase ($5' \rightarrow 3'$) & ori.



Transcription: DNA to RNA by RNA polymerase

Transcriptional Unit

- Not both strands are transcribed
- Not complete DNA is transcribed



Bacterial transcription

- Polymerase binds at Promoter
  - with σ : Initiation
  - Alone: Elongation
  - with ρ: Termination
- Bacterial genes are polycistronic.

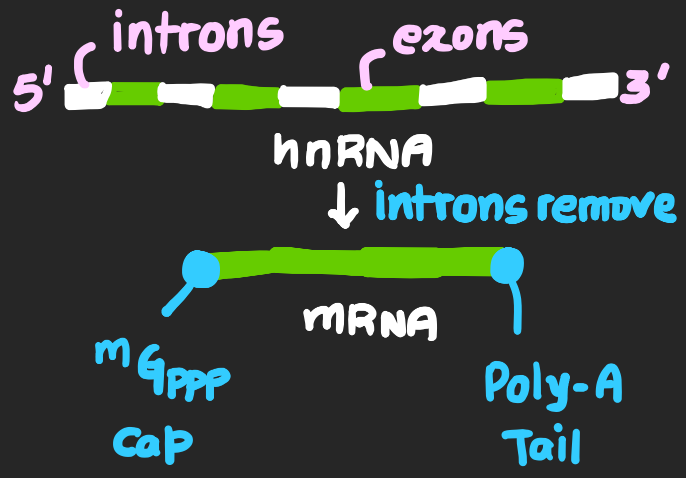
Eukaryotic Transcription

3-RNA polymerases & post transcriptional modifications

Pol·I : rRNA (28, 18, 5·8 S)

Pol·II: mRNA (hnRNA)

Pol·III: tRNA, 5S rRNA, SnRNA



Genetic codes

Triplet: by George Gamow

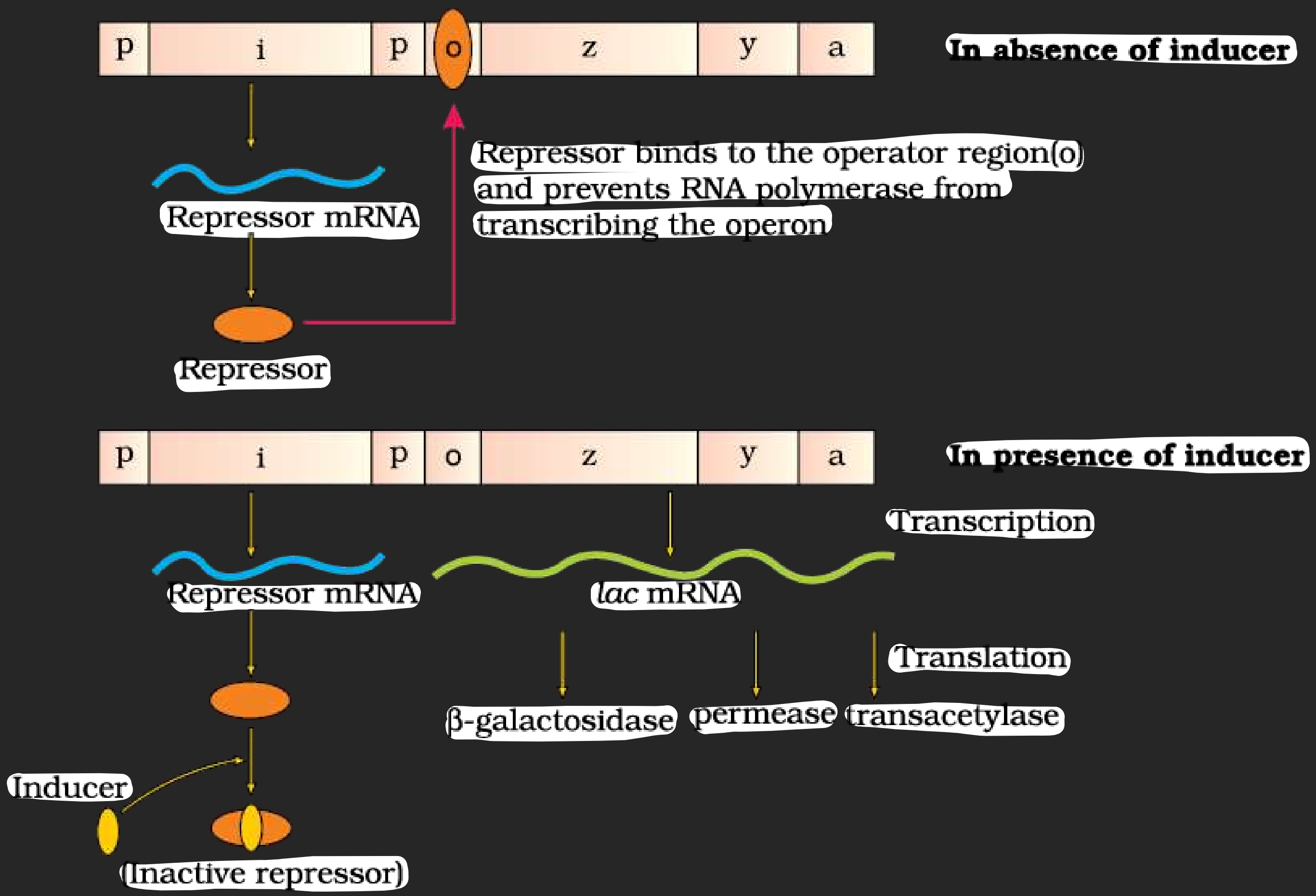
- out of 64 codons: 3 are stop codons: UAA, UGA, UAG
- AUG- start codon & methionine
- Almost universal
- Degenerate & unambiguous!

Second letter

First letter	U	C	A	G	Third letter
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Translation: RNA to protein

- tRNA charging
- charged tRNA come to AUG; small ribosomal subunit binds to mRNA
- Mg^{2+} is required to connect both subunits of ribosomes.
- Amino acids are joined one by one & Ribozyme makes bonds b/w them
- At stop codon; release factor terminates the process.



Lac operon

Lactose

Many genes regulated by Common promoter & operator.

Human Genome Project: Megaproject (9B$)

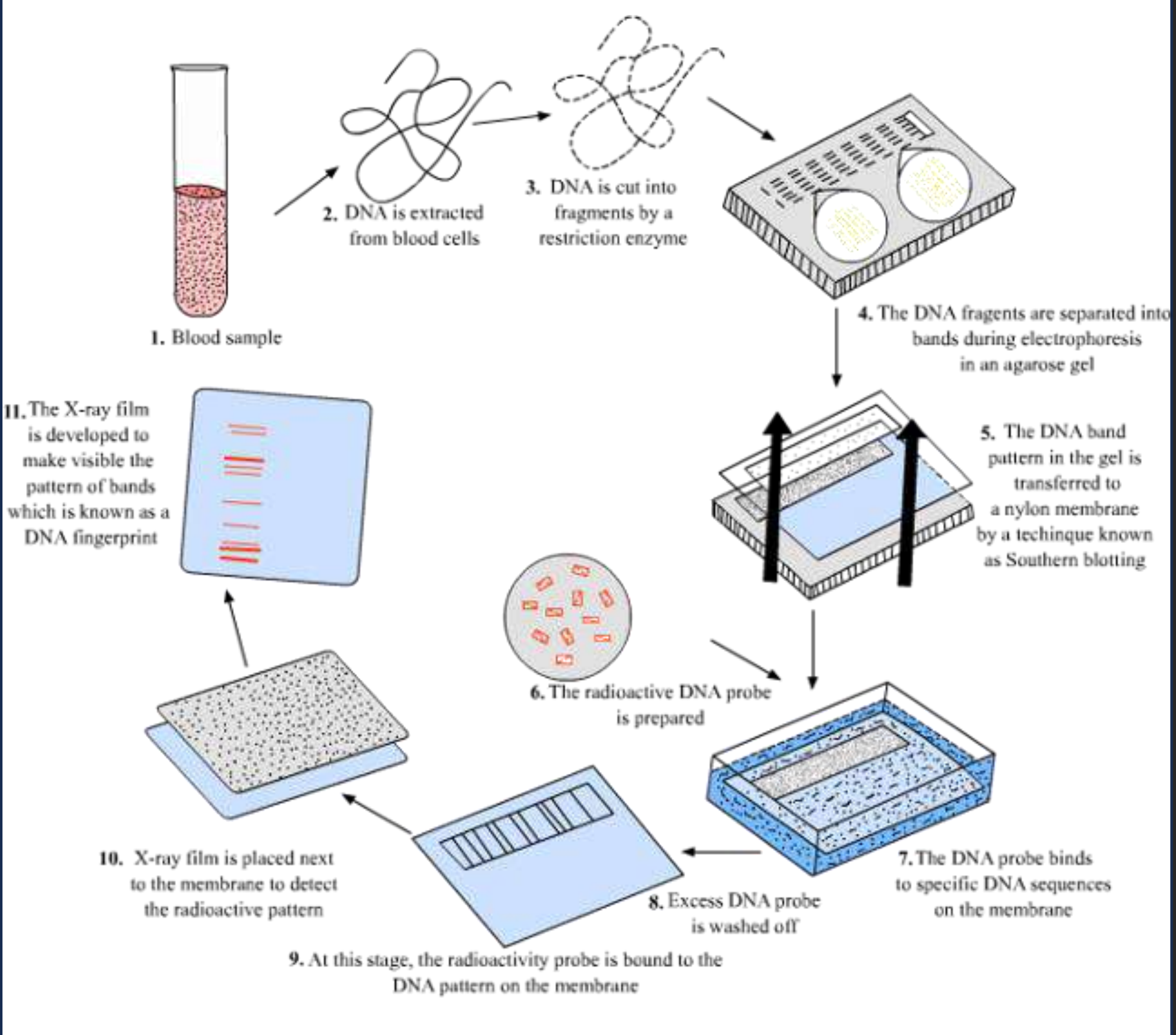
- 1990-2003
- Methodology
  - Whole genome sequencing
  - Expressed Sequence Tags

using Sanger's sequencing.

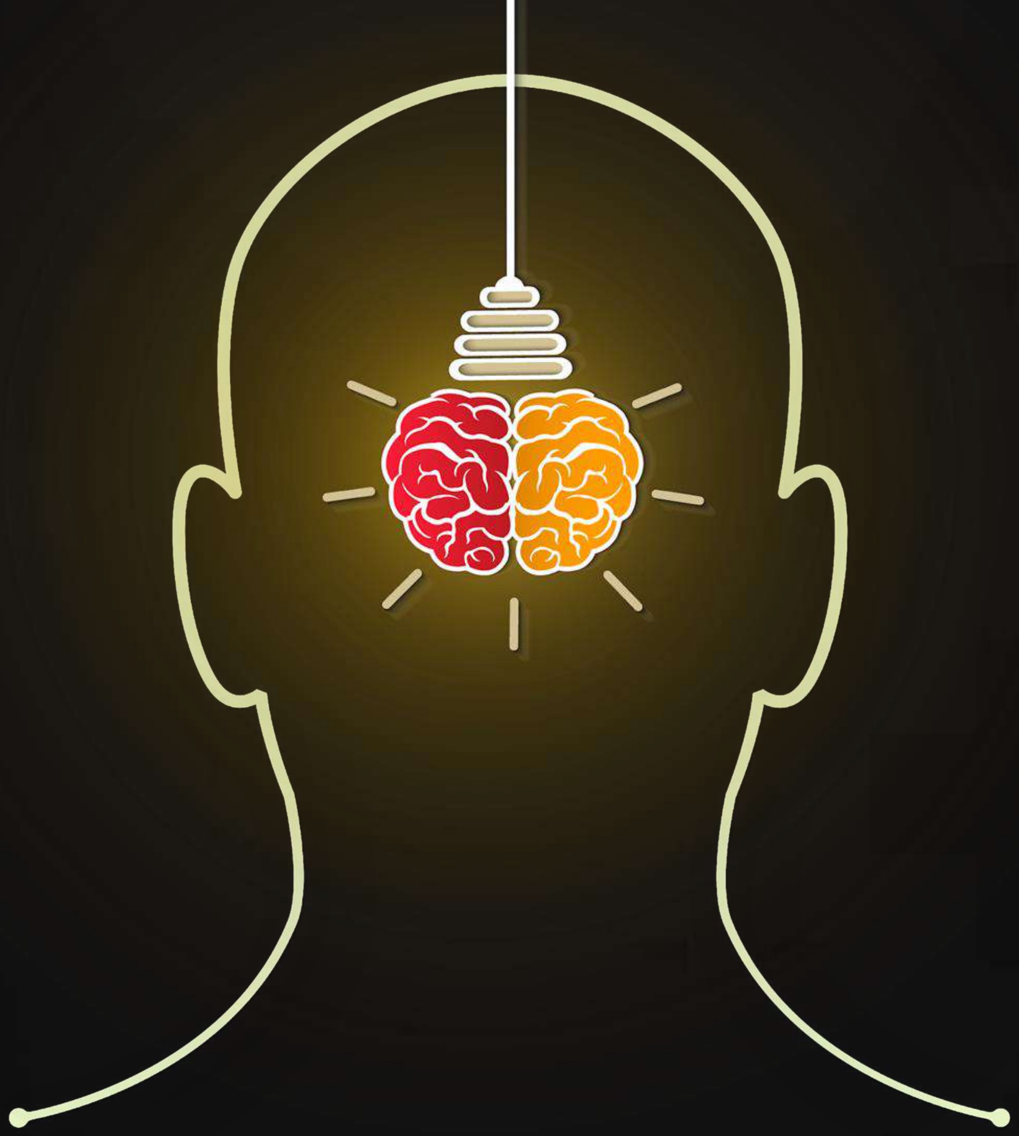
Goals	Results
• To complete sequencing of 3-billion bases & around 20-25K genes.	3·164 Billion bases ~ 30000genes (avg·size 3000bp (Dystrophin = 2·4 million bp)
• Storage, processing & retrival of data.	• < 2% content code for proteins
• Address ELSI.	• SNPs @ 14 lakh location • chr·1: 2968 genes ; Y: 231 genes.

DNA Fingerprinting : By Alec Jaffery ; based on VNTR [0·1 - 20kb size]

• Important for solving paternity & crime cases.



Steps in DNA fingerprinting techinque



THANK YOU